Kinematics of Rigid Bodies

Engineering Mechanics (ME 101)

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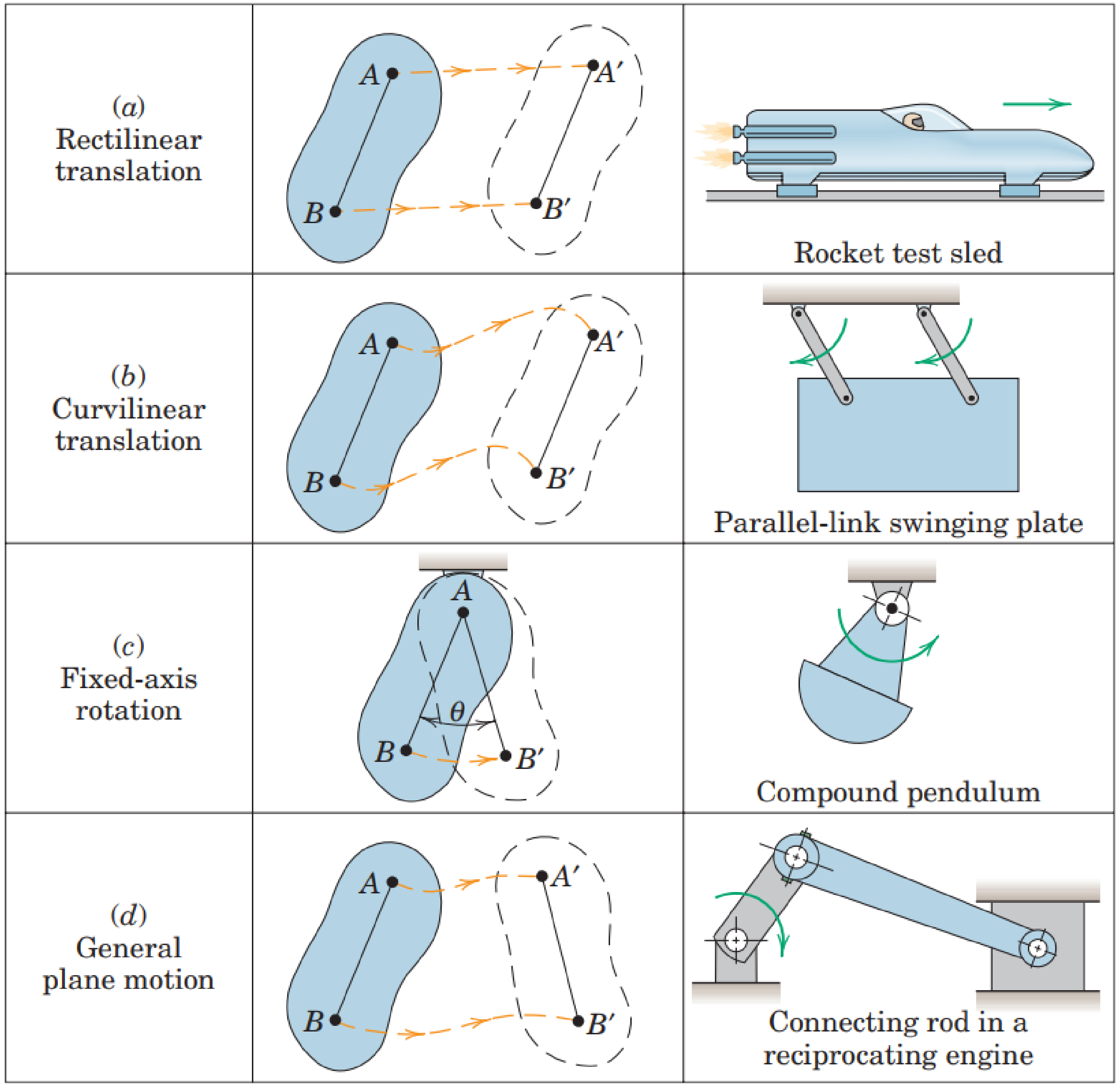
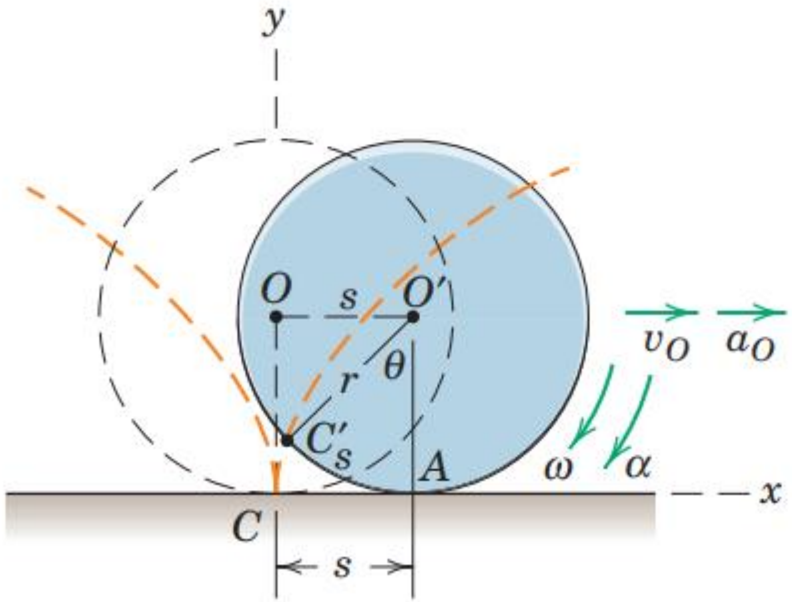
Plane motion



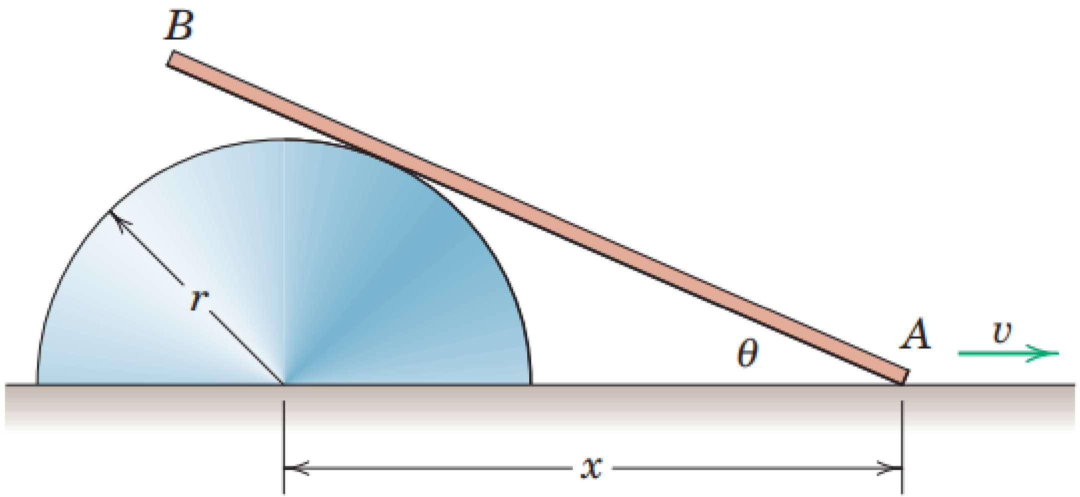
Example Problem (absolute motion)

A wheel of radius r rolls on a flat surface without slipping. Determine the angular motion of the wheel in terms of the linear motion of its center O. Also determine the acceleration of a point on the rim of the wheel as the point comes into contact with the surface on which the wheel rolls.



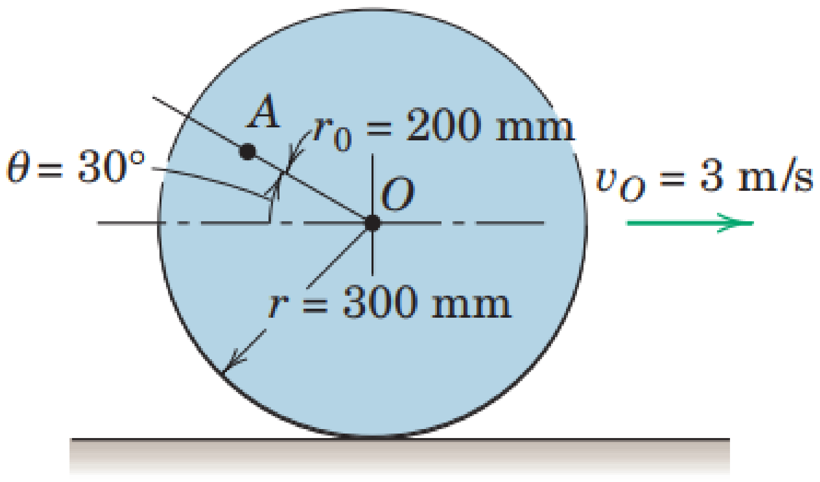
Example Problem

As end A of the slender bar is pulled to the right with the velocity v, the bar slides on the surface of the fixed half-cylinder. Determine the angular velocity $\omega = \dot{\theta}$ of the bar in terms of x.



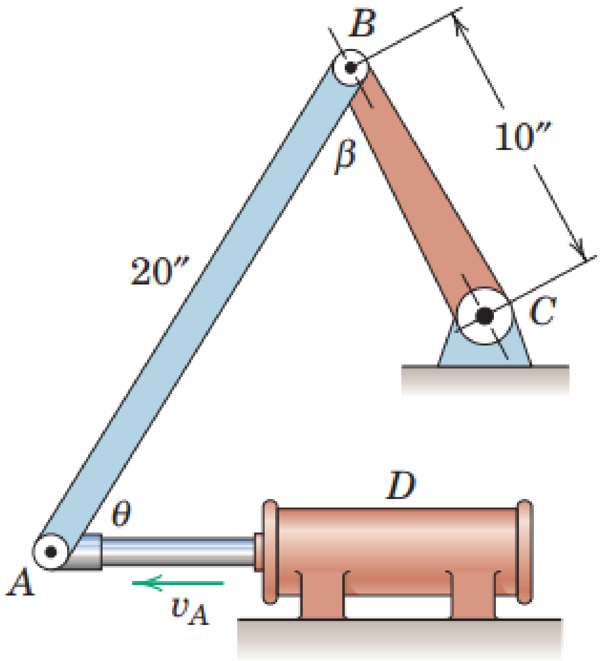
Example Problem (relative motion)

The wheel of radius $r = 300$ mm rolls to the right without slipping and has a velocity $v_O = 3$ m/s of its center O. Calculate the velocity of point A on the wheel for the instant represented.



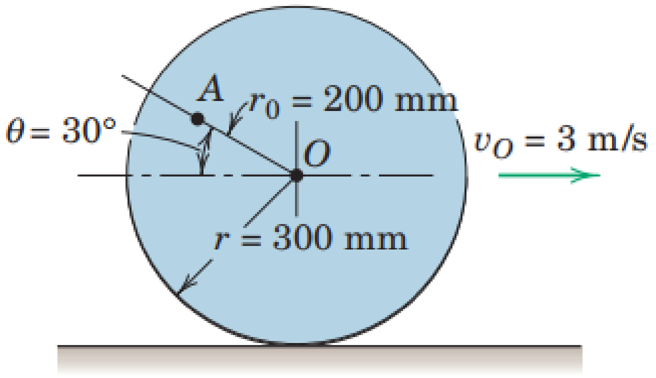
Example Problem

- For an interval of its motion the piston rod of the hydraulic cylinder has a velocity $v_A = 4$ ft/sec as shown. At a certain instant $\theta = \beta = 60°$. For this instant determine the angular velocity ω_{BC} of link BC.



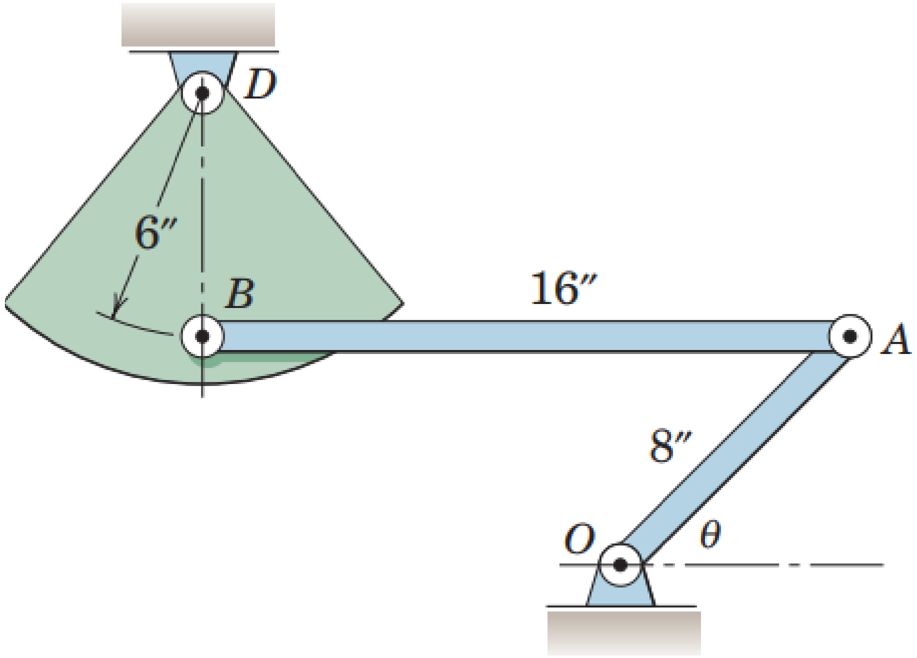
Example Problem (instantaneous center of velocity)

The wheel of Sample Problem 5/7, shown again here, rolls to the right without slipping, with its center O having a velocity $v_O = 3$ m/s. Locate the instantaneous center of zero velocity and use it to find the velocity of point A for the position indicated.



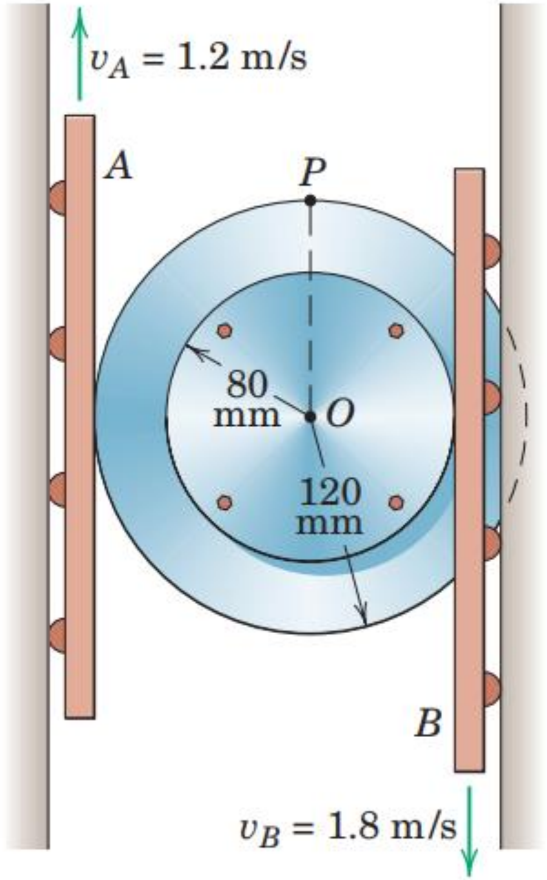
Example Problem

Link OA has a counterclockwise angular velocity $\dot{\theta} = 4$ rad/sec during an interval of its motion. Determine the angular velocity of link AB and of sector BD for $\theta = 45°$ at which instant AB is horizontal and BD is vertical.



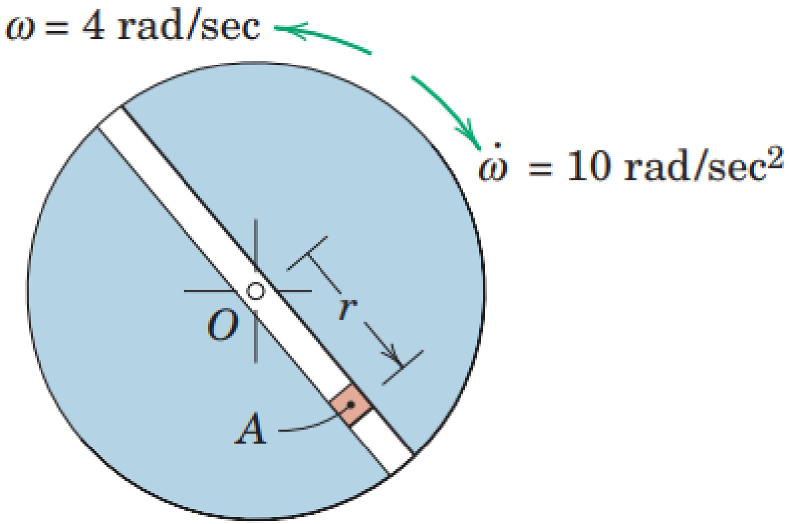
Example Problem

The sliding rails A and B engage the rims of the double wheel without slipping. For the specified velocities of A and B, determine the angular velocity ω of the wheel and the magnitude of the velocity of point P.



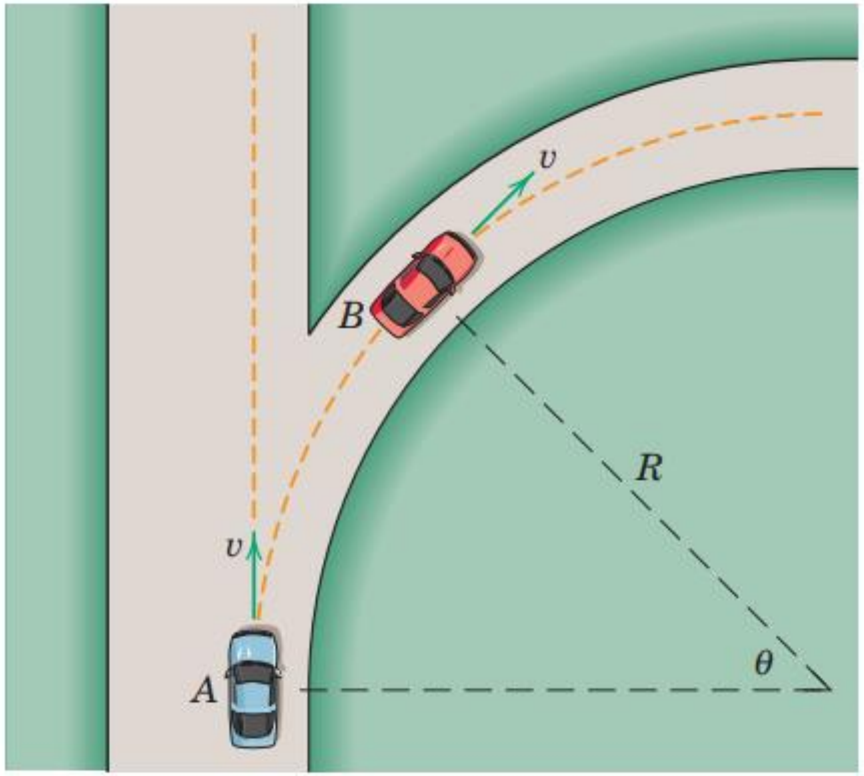
Example Problem (rotating axis)

At the instant represented, the disk with the radial slot is rotating about O with a counterclockwise angular velocity of 4 rad/sec which is decreasing at the rate of 10 rad/sec^2. The motion of slider A is separately controlled, and at this instant, r = 6 in., $\dot{r}$ = 5 in./sec, and $\ddot{r}$ = 81 in./sec^2. Determine the absolute velocity and acceleration of A for this position.



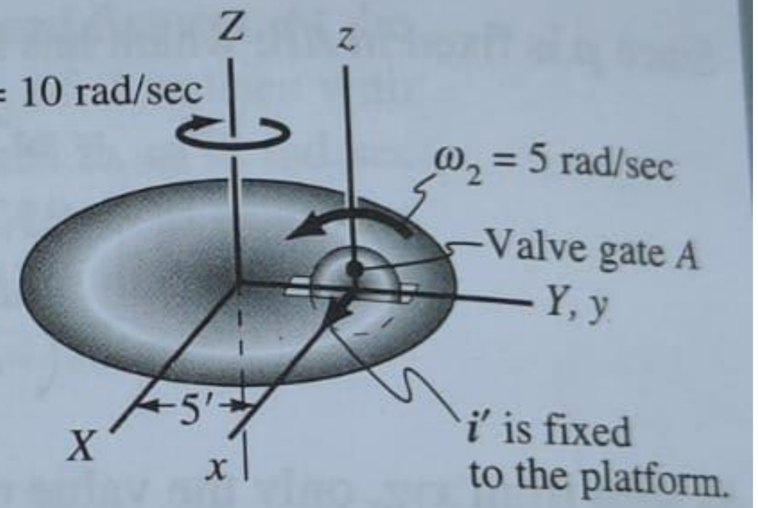
Example Problem

Car B turns onto the circular off-ramp with a speed v. Car A, traveling with the same speed v, continues in a straight line. Prove that the velocity which A appears to have to an observer riding in and turning with car B is zero when car A passes the position shown regardless of the angle θ.



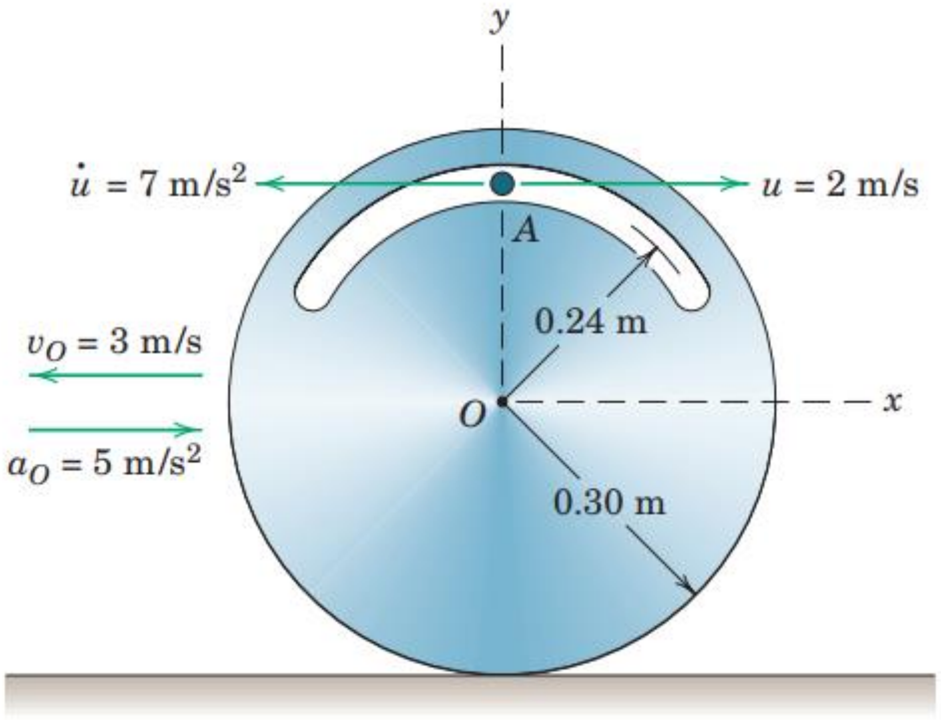
Example Problem

A wheel rotates with an angular speed ω_2 of 5 rad/sec on a platform which rotates with a speed ω_1 of 10 rad/sec relative to the ground as shown in Fig. 13.32. A valve gate A moves down the spoke of the wheel, and when the spoke is vertical the valve gate has a speed of 20 ft/sec, an acceleration of 10 ft/sec^2 along the spoke, and is 1 ft from the shaft centerline of the wheel. Compute the velocity and acceleration of the valve gate relative to the ground at this instant.



Example Problem

The disk rolls without slipping on the horizontal surface, and at the instant represented, the center O has the velocity and acceleration shown in the figure. For this instant, the particle A has the indicated speed u and time rate of change of speed $\dot{u}$, both relative to the disk. Determine the absolute velocity and acceleration of particle A.



Thank you